



Reconstructing ancient genomes from gene counts: A robust likelihood framework with sampling bias correction

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Deducing the makeup of ancient genomes is a fundamental challenge in evolutionary biology. While vast genomic datasets exist that span the entire tree of life, current methods for ancestral reconstructions struggle to resolve the inherent ambiguities of gene-sequence evolution at scale. Here, we present a numerically robust computational framework that overcomes the topological uncertainty of gene trees. Instead of tracking every single event, our phylogenetic gain-loss-duplication (GLD) model is based on birth–death processes over gene copies along the species tree. We show that the likelihood and its gradient can be computed efficiently under an adjustable observation bias of minimum gene family size. The framework facilitates unconstrained numerical likelihood maximization and ancestral inference by posterior probabilities. We apply this framework to kingdom-level reconstructions over a 269-genome archaeal dataset and demonstrate that GLD recovers ancestral states with high accuracy. We compare GLD inference with phylogenetic reconciliation from gene sequences (ALE method) and show that implausibly frequent horizontal gene transfer inferred by ALE are often statistical artifacts of collapsing phylogenetic signals in large alignments. In contrast, GLD inferences reveal how two layers of opposing evolutionary mechanics shape microbial genomes: a high-frequency tension between genome streamlining and the pervasive influx of transient genes, complemented by an adaptive counterbalance of recurrent, modular losses, and punctuated massive gains. The GLD framework provides a statistically sound foundation for hypothesizing about gene content evolution across the diversity of entire kingdoms.

evolutionary genomics | ancestral reconstruction | birth–death process

Prokaryotic genome content is shaped by vertical evolutionary events of inheritance, loss, and duplication, as well as by the nonvertical gain of genes (1). As a result, genes with common ancestry can be grouped into homolog families across extant genomes and the families can be delineated by the similarities between gene sequences (2–4). Datasets of homolog families are the foundational basis for inference in evolutionary genomics.

For a mathematical abstraction of genome evolution, we can imagine the gene families as populations within the genome, affected by birth (duplication, horizontal transfer, or origination) and death (loss) events over family members as they are inherited along a species phylogeny. A corresponding stochastic model is a continuous-time linear birth–death–immigration (BDI) process that treats every gene family as a population of identical and independently evolving copies. The process is linear because duplication (birth) and loss (death) events occur to every copy with the same intensities; the immigration component accounts for gains occurring independently from current copies. Linear BDI processes are computationally advantageous because they have well-characterized transient distributions (5). Here, we study inference with a probabilistic gain-loss-duplication (GLD) model based on BDI processes running on the edges of a species phylogeny (6, 7).

We compare inference from gene copies to an alternative, but computationally more costly analysis method for genome evolution called phylogenetic reconciliation (8), which takes gene sequences into account. Reconciliation frameworks combine a molecular sequence evolution model with a duplication-transfer-loss (DTL) model over distinguishable copies. Inferring the residue-level homologies (multiple alignments) together with the DTL events is quite difficult because they depend on each other (9). Given the combinatorial and computational complexities of multiple alignment and phylogeny construction, bioinformatics algorithms use various heuristics to avoid superexponential blow-up, and typically reconcile gene phylogenies built from fixed multiple sequence alignments (10–13).

In contrast, inference from gene copies (14–16) depends solely on how many representatives of a given family are present in each sampled genome. The corresponding

Significance

How has evolution shaped the diverse gene repertoires of extant genomes? We find that current methods seeking to reconcile a genome phylogeny with complex gene sequence histories quickly hit a crisis-point where the phylogenetic signal for gene families vanishes as the dataset expands. We introduce a numerically robust inference framework for the gain-loss-duplication (GLD) model that tracks gene-copy evolution without requiring resolved gene trees. We demonstrate that the GLD framework provides a scalable inference method for reconstructing ancient genomes and a solid foundation for analyzing evolutionary dynamics. Our experiments reveal that archaeal evolution is characterized by a dynamic equilibrium of rare massive gene gains and modular, recurrent losses.

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input data collection is cheap because pairwise alignments suffice to group the sequences into homolog families or to classify genes by a reference set of ortholog groups (2–4). Moreover, copy gains do not require a determined origin, eliminating the need for gene trees (either explicitly during reconciliation or implicitly as a guide tree to multiple alignment).

The formal phylogenetic (GLD) model (6, 7) defines the joint distribution of node copies ξ_v (nonnegative integer random variables) associated with the nodes v of a phylogeny T . Copy evolution on each edge $uv \in T$ follows a continuous-time BDI process $\{\xi(t) | 0 \leq t \leq t_v\}$ that starts with $\xi(0) = \xi_u$ copies at the parent (u), and arrives to $\xi(t_v) = \xi_v$ copies at the child (v) over some time t_v (the edge length). The process on each edge is parameterized by the instantaneous rates of birth, death, and immigration. Birth and death events occur to each copy independently, corresponding to duplications and losses. Immigration events occur independently of existing copies, modeling gains by horizontal transfer or origination. The equivalent Kolmogorov backward equations for edge uv are

$$\frac{\partial \mathbb{P}\{\xi(t) = 0\}}{\partial t} = \mu_v \mathbb{P}\{\xi(t) = 1\} - \gamma_v \mathbb{P}\{\xi(t) = 0\}$$

and for

$$0 < n, \frac{\partial \mathbb{P}\{\xi(t) = n\}}{\partial t} = (n+1)\mu_v \times \mathbb{P}\{\xi(t) = n+1\} + (\gamma_v + (n-1)\lambda_v) \times \mathbb{P}\{\xi(t) = n-1\} - (\gamma_v + n\lambda_v + n\mu_v) \times \mathbb{P}\{\xi(t) = n\}.$$

We scale edge lengths and rates so that loss rate $\mu_v = 1$ at every node v . The model's free parameters are the edge length and the (loss-relative) duplication and gain rates (t_v, λ_v, γ_v) at every nonroot node v , as well as (λ_R, γ_R) at the root R . The copies across the extant genomes form the observable profile for each family. Given the set of homolog family profiles, we want to infer the model parameters and the copies in ancestral genomes.

We have simple closed formulas for the BDI process transition probabilities (*Theorem 1*), which can be employed in recurrences for various probabilities conditioned on ancestral copies. Unfortunately, recurrences that condition on the hidden variables involve infinite summations because of the theoretically unbounded ancestral copies that include the genes that disappeared without a trace.

It is possible to compute the probabilities by conditioning on surviving copies that are progenitors of extant genes (7). The likelihood, posterior, and gradient calculations can now be factorized by conditioning on the values of surviving copies (*Theorem 2*) which are necessarily (by the pigeonhole principle) at most as large as the copy sum across extant genomes. We developed a numerically stable exact gradient computation algorithm that works with generic unconstrained nonlinear optimization procedures.

Importantly, our calculations (*Theorems 3–5*) account for inherent sampling bias in ortholog datasets, defined as the minimum number $\Omega_{\min}$ of copies in observed gene families. An obvious bias is that families are known only if at least $\Omega_{\min} = 1$ member is identified. Conservatively, we may want at least 2 members to exclude orphans which were annotated by a de novo gene finder but have no evidence of function by sequence alignment. Gene-tree reconciliation methods need at least 4 copies, since fewer have trivial unrooted gene phylogenies.

We demonstrate the performance of our methods on a dataset covering 269 complete genomes across Archaea. Our

kingdom-level reconstructions reveal the prevalence of genome streamlining, the recurrence of modular loss in unrelated lineages, and a pangenome-level equilibrium maintained by rare genes. Additionally, we compare the ancestral reconstructions from gene copies and by phylogenetic reconciliation from gene sequences (the ALE method) side-by-side over the three datasets across 30 plants, 60 archaea, and 265 bacteria (17–19). For the two studies we find similarly distributed ancestral estimates. Over bacteria, however, the GLD reconstruction infers a loss-dominated scenario in contrast to the ALE reconciliation that postulates an extraordinarily high frequency of gene transfers and a very small root genome. We observed that the large alignments underlying the reconciliation have insufficient phylogenetic signal; thus, gene trees are resolved by essentially randomly placed transfers. We observed that two other ALE reconciliation studies (20, 21) over 700 and 1,007 genomes also posit rampant transfer and small genomes for the deepest ancestors. We suggest that reconciliation methods are pushed beyond their limits at such scales: gene trees for large families are mathematically underdetermined because of bounded gene length.

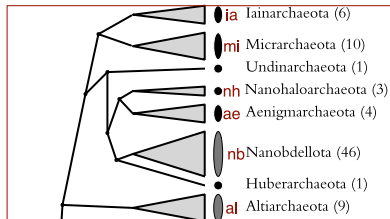
Results

We developed numerically robust algorithms for gradient computation in phylogenetic GLD models starting from the mathematical framework of Csürös (7). In particular, the recurrences for likelihood, posterior, and gradient calculations had to be carefully designed to attain numerical stability (*SI Appendix, section A*). The key to robustness and well-scaled derivatives is to employ logit-scaled distribution parameters with the biologically inspired restriction that duplication and gain rates are bounded by the loss rate. Furthermore, we developed a fast algorithm for arbitrary observation bias correction (*Theorems 3–5*). We implemented the algorithms in conjunction with numerical optimization by the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm (22) in Java as part of the Count software package (23). Requiring quadratic instead of cubic time in function of tree size, Count is not only more precise and flexible but also faster than older versions (16), so that it scales well for inference with hundreds of genomes.

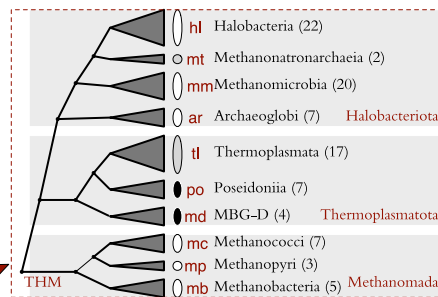
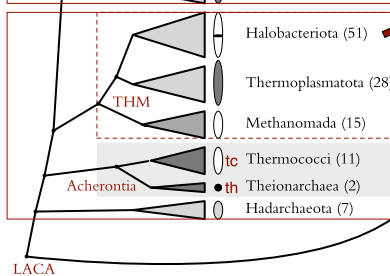
We compiled a collection of homolog families across 269 Archaeal genomes from ref. 24, pinned to the arCOG database of Archaeal Clusters of Orthologous Groups (25) and to the asCOG database of Asgard clusters (26). We evaluated GLD inference along the phylogeny shown in Fig. 1, after a recent suggestion by Baker et al. (27) for the placement of Nanobdellati within Methanobacteriati. The experiments use subsets of 114, 80, and 75 and 194 genomes over Methanobacteriati (Euryarchaeota), Nanobdellati (DPANN group), Proteoarchaeota (Asgardarchaeota and the TACK group), and Methanobacteriati with Nanobdellati. We denote the four genome sets by E114, D80, P75, and ED194.

Algorithm Adequacy: Convergence and Predictive Power. We evaluated the convergence of the likelihood optimization separately on the four datasets, retaining only the families with at least four members in the selected genomes. We found that the optimization consistently converged to a stable maximum in several thousand iterations (*SI Appendix, section B.1*). Most of the time is spent on gradient calculation, which requires only a few seconds per evaluation on a standard laptop. Notably, nearly all BFGS iterations required only a single gradient evaluation, demonstrating that the local curvature is reliably approximated

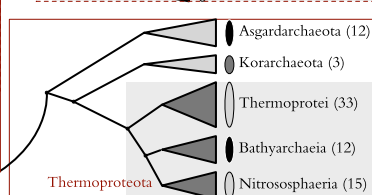
Nanobdellati (80)



Methanobacteriati (114)



Proteoarchaeota (75)



- all MAGs
- mostly MAGs
- partly MAGs
- 1 MAG
- no MAG

Fig. 1. Assumed phylogeny of 269 Archaea with number of genomes between parentheses and two-letter codes for taxonomic groups. Light triangles denote phyla, dark triangles denote classes and superclasses. Shaded ovals indicate metagenome-assembled genomes (MAGs). The *Inset* on the *Upper Right* side shows the relationships between the classes in the THM group formed by Thermoplasmata, Halobacteriota, and Methanomada.

by the reverse Jacobian recurrence in our numerically stable implementation.

The optimized models are used to compute posterior statistics for ancestral gene repertoires. We assessed reconstruction fidelity of two primary predictors: the maximum a posteriori (MAP) estimate for family presence (thresholded at $P \geq 0.5$) and the posterior expected number of families. In order to quantify the identifiability of the phylogenetic GLD models, we performed discrete-event simulation of the BDI processes using empirical rates inferred from data, and compared the posterior estimators to the true values known from the simulation. While previous benchmarks for GLD inference were limited to small-scale trees of six genomes (28), our analysis extends this to kingdom-level phylogenies.

Fig. 2A illustrates the reconstruction accuracy for simulated families in the ED194 dataset (Methanobacteriati and Nanobdellati). The posterior estimates for total ancestral families have remarkable precision throughout the tree. For MAP family presence, false positives (FP) and false negatives (FN) remain low, ranging from a few dozen families at most nodes to approximately 100 at the deepest ancestors. When viewed as a family classification problem (Fig. 2B), the MAP estimator maintains a negligible false positive rate (maximum 0.8%, typically $< 0.2\%$) while recovering nearly the entire ancestral genome content (median TPR 98.6%). This accuracy closely approaches the theoretical limit of information available in numerical profiles (Fig. 2E). We observe a clear scaling effect with sample size: A model optimized on about a thousand families (which is close to the number of model parameters) remains robust (Fig. 2C), but a hundred families are insufficient for reliable estimation at many ancestors, particularly within Nanobdellati (Fig. 2D). As expected, the root genome represents the most difficult challenge, even when the true model is known. Nevertheless, posterior reconstruction under the GLD model significantly outperforms simpler alternatives: both probabilistic presence-absence models [e.g., GLOOME (29)] and numerical parsimony exhibit substantially higher error rates and lower TPR. The same high accuracy is observed across other kingdom-level datasets (P75, D80, and E114), as well as in smaller benchmarks over phyla (SI Appendix, section B.2).

These results demonstrate that our algorithmic framework ensures fast and accurate inference in the GLD model.

Model Adequacy: Genome Size and Family Frequency Distributions. We compared the distribution of family profiles in simulated data to the observed distribution across the four kingdom-level genome sets. Fig. 3 shows the distribution of the number of families at tree nodes in ED194. The GLD model provides an excellent fit for genome size scaling: across all taxa, 62.8% fall within the interquartile range of 1,000 simulations, and 98.7% fall within the 95% CI. The phylogenetic GLD model thus explains the joint distribution of genome sizes across multiple phyla (see SI Appendix, section B.3 for other datasets).

Family frequencies in extant genomes, however, deviate from the GLD model (SI Appendix, section B.3). As characterized in previous studies (1, 25, 30), the frequency distribution consists of a small core of nearly universal families, a larger shell, and a vast cloud of rare families (Fig. 4). Our technical definition is that the components cover consecutive thirds of the frequency range. We define a fourth “halo” component that comprises excluded families below sampling bias. Assuming that the vast majority of profiles are shaped mainly by genetic drift (31), the phylogenetic GLD model serves as a null model to detect nonneutral patterns. The graph of cumulative residuals (SI Appendix, Fig. S27) reveals three regions of directional evolutionary forces, separated by points where the model’s and the dataset’s distribution functions closely agree. In the left (transient) region, family frequencies shift to the left with a maximum difference near the observation bias, due to the unexpectedly high occurrence of transient families. The intensity of the transient force is manifested also in the halo families that outnumber included families by six to one (exceeding twelvefold the model expectation of one-to-two). The adjustable sampling bias mitigates the transient effect and thus amplifies the phylogenomic signal for ancestral reconstruction. In the middle (adaptive) region, frequencies also shift to the left. We argue in the next section that such families undergo recurrent and modular losses that are underestimated by the GLD model. Frequencies in the right (stabilizing) force region are shifted to the right because of unexpectedly strong conservation. The breakpoints between the force regions are

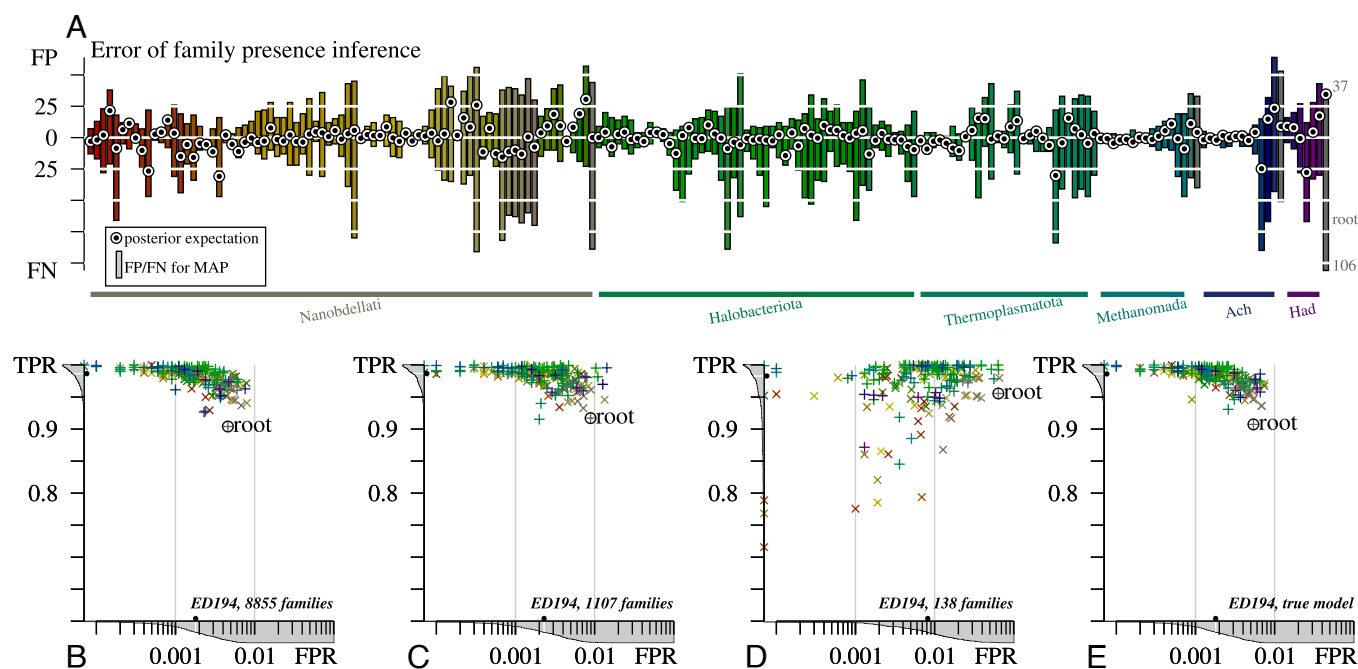


Fig. 2. Error of ancestral family reconstruction on simulated data over ED194. (A) Upward and downward bars show FP and FN for the MAP estimates of family presence at ancestral nodes. Nodes are enumerated in postorder and colored by position in the phylogeny as in Fig. 3. Circles show the error of the posterior expected family count. (Ach: Acherontia; Had: Hadarchaeota.) (B–D) Linear-log plot of false positive rate (FPR) and true positive rate (TPR) by the MAP estimate at ancestral nodes, with decreasing sample sizes. The model is optimized on all families (B), or on one-eighth (C), or on one-sixty-fourth (D) as many families. (E) FPR and TPR for the posterior inference with the true model (the Bayes classifier). On B–E, Methanobacteriati are plotted with + and Nanobdellati with ×; shaded graphs on the exterior of the axes plot the cumulative distribution function across ancestral nodes, and dots mark the median TPR and FPR. Statistics are computed on an independent sample of 10,000 simulated families.

at $\sim 1/6$ and $\sim 45\%$ of the range, while the breakpoints between intervals where the model over- and underestimates profile are near the shell boundaries at $\sim 1/3$ and $\sim 2/3$ of the range (*SI Appendix, Table S1*). These results suggest that deviations of family frequency uncover biological forces that populate the cloud, deplete the shell, and maintain the core.

Model Deviations: Expanding Ancestors and Cryptic Inheritance. The phylogenetic GLD model can accommodate conserved core profiles by introducing rate categories that model strong purifying selection. Shell families, however, exhibit complex histories, likely due to lineage-specific selection and functional dependencies between genes. To investigate these

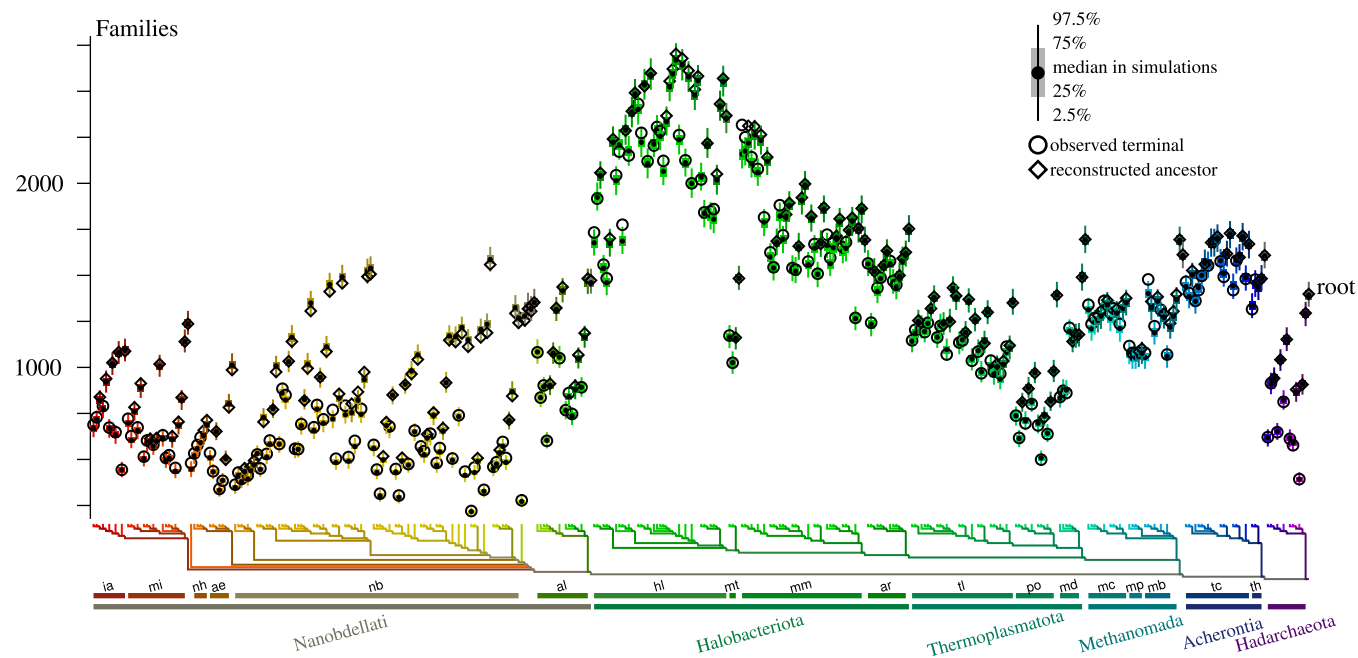


Fig. 3. Posterior expected family counts at ancestral nodes (diamonds), observed family counts at terminal nodes (circles), and empirical distribution of family counts in 1,000 simulated datasets (box-and-whisker plot with 95% CI). The small interquartile ranges show that genome size distribution is highly concentrated in the GLD model. Phylogeny nodes are enumerated and colored as shown by the tree diagram along the X axis. Two-letter codes denote the major clades with at least two genomes from Fig. 1.

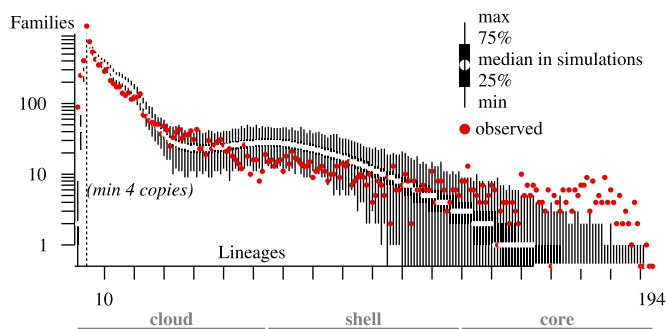


Fig. 4. Distribution of family frequency (extant genomes with a copy) in simulated and in observed profiles over ED194; on linear-log scale. Box-and-whisker plot shows the distribution of families with given frequency in simulations; red discs plot the observed counts in ortholog data. The cloud, shell, and core families fall into consecutive thirds of the frequency range.

dynamics, we examined clade-specific variations and the impact of taxonomic context on ancestral inference.

First, we examined profile statistics for different, well-represented clades at different taxonomic levels in simulations and empirical datasets (*SI Appendix, section B.4*). Simulated data produce significantly more unique profiles ($P < 0.001$ with one exception in P75 and E114), which is consistent with independent family evolution. At the same time, empirical datasets tend to have significantly more clade-specific families of at least four copies (with $P < 0.01$ at about 50% of clades, excluding the root and its children). The two tendencies lead to higher recurrence of clade-specific profiles, suggesting that gene families evolve by constraints that maintain functional repertoires within clades.

Second, we examined how ancestral inference changes in increasingly larger phylogenetic contexts. For example, within Methanobacteriati, we inspected the ancestors of the main three groups Thermoplasmata, Halobacteriota, and Methanomada across datasets of increasing scale (*Table 1*). Paradoxically, ancestral estimates increase as more taxa are included (also in P75: *SI Appendix, Table S6*). In simulations, ancestral estimates remain stable regardless of context (*SI Appendix, Table S5*), implying that the expansion observed in empirical data is a biological signal rather than a statistical artifact.

For families with shifting inference, the larger dataset may uncover “cryptic” inheritance, when lineage-specific gains in smaller dataset are revealed as ancient vertical inheritance. This is contrasted with “fake” inheritance (lineage-specific gains in larger dataset), a technical artifact caused by placing the root too close to a child node. For example, for M15 (Methanomada) and H51 (Halobacteriota), families that are specific to the root’s closest child (Methanococci and Archaeoglobi) are inferred on the smaller dataset as losses in the sister branch.

Cryptically inherited genes typically belong to the shell or cloud in restricted datasets but are discovered as core components in larger contexts. In Thermoplasmata, cryptically inherited genes appear only in one of the classes but are well represented in Halobacteriota and Methanomada. The functional characterization shows that about half of cryptically inherited genes are metabolism-related and that they often constitute sets of genes on the same pathway (*SI Appendix, section B.5*). Particularly noticeable is the high number of genes involved in the metabolism and transport of amino acids and coenzymes. Namely, many genes for amino acid synthesis are missing in the marine lineages (*32, 33*) but present in the class of Thermoplasmata: Cryptically inherited genes include sets for the biosynthesis of isoleucine and valine, of

arginine, of histidine, and of aromatic amino acids. Cryptically inherited genes for coenzyme metabolism are typically found only in Methanomassiliococcales among Thermoplasmata, and include the sets for the biosynthesis of cobalamine (13 families), siroheme (6 families), and F430 coenzyme (4 families) that are involved in vitamin B12 metabolism (*34*), as well as the subunits for the alkyl-coenzyme M reductase (MCR) complex that are found in methanogens (*35*). Cryptically inherited families in Halobacteriota include the MCR families, as well as the subunits of the tetrahydromethanopterin S-methyltransferase complex (MtrA–G) that joins MCR to the Wood–Ljungdahl pathway in methanogens (*35, 36*). The corresponding genes are absent in the mostly nonmethanogenic Archaeoglobi (*37*).

These results show that GLD inference benefits from expanding phylogenetic contexts that allow for discovering inheritances otherwise masked by modular loss.

Methodological Differences: Phylogenetic Reconciliation and Inference from Copy Numbers.

Previous comparisons between GLD reconstruction and the ALE reconciliation method (2×28 fungi and 40 cyanobacteria) reported comparable dynamics and a close agreement on (cloud) families with up to 8 copies. We expanded this comparison to three published duplication-transfer-loss (DTL) reconciliations by ALE over varying scales (*17–19*). We find that the methods converge on small datasets but fundamentally disagree on large ones (*SI Appendix, section B.6*).

On a dataset of 30 genomes of plants and green algae (*18*), the reconstructions are highly consistent (*SI Appendix, Fig. S30*). Both methods agree on the size of the common ancestor of land plants and the root’s approximately 5,000 genes (as shared inheritance in GLD). Similarly, on a 60-genome archaeal dataset, the GLD and DTL reconstructions of ancestral copies are comparable, although DTL tends to infer larger estimates at specific nodes (*SI Appendix, Fig. S32*). However, DTL infers a much smaller Last Archaeal Common Ancestor (LACA) with only ~ 400 genes (which is fewer than the smallest extant genome in the dataset), missing many key genes from main metabolic pathways (*17*).

Using ALE across 265 bacterial genomes, Coleman et al. (*19*) inferred abundant horizontal transfer, suggesting that more than 80% of genes across all phyla were acquired since the phylum’s radiation. Three reconstructions with different species trees trace a few hundred families back to the deepest ancestors and only 40 to 51 to the last bacterial common ancestor (LBCA). In contrast, the GLD model infers much larger ancestors (*SI Appendix, Fig. S33*). Even when restricted to a functionally annotated dataset over COG (clusters of orthologous genes) domains (*2*), GLD estimates remain consistently above DTL inferences (*SI Appendix, Fig. S34*). These results indicate that ALE systematically favors recent gene origins by HGT over vertical inheritance, placing implausibly few genes at the deepest ancestors. As we show in the following section, the reason is not simply a modeling trade-off (e.g., of high transfer rates), but the consequence of information decay in sequence alignments.

Critical Assessment: Large-Scale Reconciliation Is Limited by Vanishing Phylogenetic Signal.

The ALE reconciliations in ref. *19* infer an extraordinary amount of horizontal transfer for shell and core families, often hundreds of events per family, while identifying hardly any duplication events (*Fig. 5 D and E* and *SI Appendix, B.7*). The number of transfers is in near-perfect linear correlation with family size ($R^2 = 0.93$). The examination of the underlying multiple alignments reveals a fundamental scaling failure caused by the limited length of gene sequences. Alignments

Table 1. Top: Reconstructed ancestral genome sizes in different phylogenetic contexts (min. 4 copies): posterior expected counts for families (*F*) and Bottom: Shared inheritance ($\hat{F}$) of root genome by both children

Dataset: Ancestor	M15/T28/H51/N46 <i>F</i>	T94 <i>F</i>	E114/D80 <i>F</i>	ED194 <i>F</i>
M15	1,266	1,304	1,336	1,397
T28	997	1,328	1,379	1,489
H51	1,588	1,462	1,609	1,753
T94		1,611	1,557	1,694
N46	388		591	1,242
D80			1,192	1,472
Methanobacteriati			1,441	1,398
Shared inheritance		$\hat{F}$	$\hat{F}$	$\hat{F}$
THM		1,139		
Nanobdellati			619	
Methanobacteriati			1,117	1,175

M15:Methanomada, T28:Thermoplasmatota, H51:Halobacteriota, T94:THM, N46:Nanobdellota.

have typically a few hundred columns (219 on average), and even fewer parsimony-informative sites (PIS; 114 on average). (A PIS is a column that has at least two residues such that each appear in at least two sequences.) Informative sites do not increase with more sequences in the alignment once a family exceeds about 10 members (Fig. 5 A and B). Consequently, the relative phylogenetic information decays as the family size grows (Fig. 5C). In particular, as alignment size (n) increases, the number of internal branches to be resolved ($n - 3$) quickly exceeds the number of available informative sites. Adding a new sequence multiplies the tree space size by $(2n - 3)$, which is the number of possible placements for the new sequence, but provides a number of informative characters that is bounded by some constant length L depending on the family. After some threshold n^* (for which L is sufficient to correctly choose among $(2n^* - 3)$ possibilities), if $n^* < n$, then each new gene is likely misplaced, and errors increase in a close correlation with n . Notably, IQ-TREE, which is the gene-tree inference engine used in ref. 19, is benchmarked on alignments where the number of sites is at least twice the number of sequences (38). The PIS/ n ratio is particularly low for the shell and core families that are essential for ancestral reconstruction. The vanishing phylogenetic signal in these families leads to high topological ambiguity, which is artificially resolved by invoking spurious HGT events (Fig. 5E). We simulated gene-tree sampling with IQ-TREE for shell and core families to confirm that as the ratio of informative sites drops (SI Appendix, Table S8), topological noise explodes with up to 80% of internal edges unrecovered from typical alignment lengths.

We find the effect of vanishing signal consistently across other large-scale ALE studies. In a reconstruction involving 1,007 genomes (20), the inferred HGT events are 25% more frequent than the family size ($R^2 = 0.96$), and duplications are very rare (SI Appendix Figs. S39 and S40). The LBCA root is reduced to 5 genes, and 20% of ancestral nodes have less than 100 genes. Similarly, a study of 700 genomes (21) shows a strong correlation between transfers family size ($R^2 = 0.83$), resulting in a last universal common ancestor (LUCA) with only 45 genes (SI Appendix, Fig. S41).

These findings confirm that sequence lengths impose a hard limit on large-scale reconciliation.

Analysis of Evolutionary Trends: Loss-Dominated Dynamics in Archaea. The GLD reconstruction over ED194 posits a root genome with a complexity similar to extant medium-sized Methanobacteriati. The gene repertoire expands until

the common parent of the Thermoplasmatota-Halobacteriota-Methanomada (THM) group and Nanobdellati (Fig. 3).

The inferred losses represent a combination of genuine evolutionary events such as genome streamlining in symbiotic Nanobdellati and technological artifacts of missing gene annotations. Nanobdellati genomes, which are primarily metagenome-derived, show loss of 28 to 36% families, compared to a median of 12 to 13% among E114 terminal lineages, which are well represented in complete genome assemblies. Crucially, incomplete genome coverage makes no difference in the GLD model because it gets perfectly absorbed by increased loss and adjusted duplication rates (7). (Consequently, genome completeness is nonidentifiable from copy numbers.)

Events over surviving copies aggregate multiple evolutionary events: a surviving-copy loss entails nonextinction in the sibling subtree and includes all histories with multiple duplications and losses resulting in extinction at all extant descendants. The shared inheritance are the surviving copies inherited by both children at a node. Shared inheritance ranges from 50% at the Nanobdellati root to 70 to 80% at THM and Methanobacteriati (Table 1). Unlike reconciliation, which would always interpret asymmetric inheritance as origination in a child lineage, GLD inference also quantifies differential loss at the root.

The analysis of family-level events in function of family frequency highlights the dominant role of loss in shaping shell and core profiles. We find that families in the middle 2/3 of the frequency range are explained by 3 to 4 family gains and an average of ~30 family losses (SI Appendix, Fig. S42). For families originating at the THM ancestor, a median of 36 to 44 extant lineages inherit a copy (out of 94, in E114 and ED194 reconstructions), and the middle half of THM-originated families appear in the shell (SI Appendix, Fig. S43). As a consequence of the combinatorics of subtrees, the distribution of independent acquisitions for a family displays geometric decay (SI Appendix, Fig. S45).

Finally, we inspected the balance of family events on different tree edges (Fig. 6). If we reconstruct the histories from samples with observation bias $\Omega_{\min} = 4$, 88% of lineages show net family loss (SI Appendix, Fig. S49). This background of gradual loss is interrupted by punctuated gains at specific milestones, such as at the ancestors of Methanosarcina, Halobacteria, Marsarchaeales, and Woese- and Pacearchaeales. Interestingly, the same clades were found to harbor integrons which facilitate gene transfers (39). When halo families are included, the dynamics shift toward a general equilibrium, with 57% of lineages showing net loss.

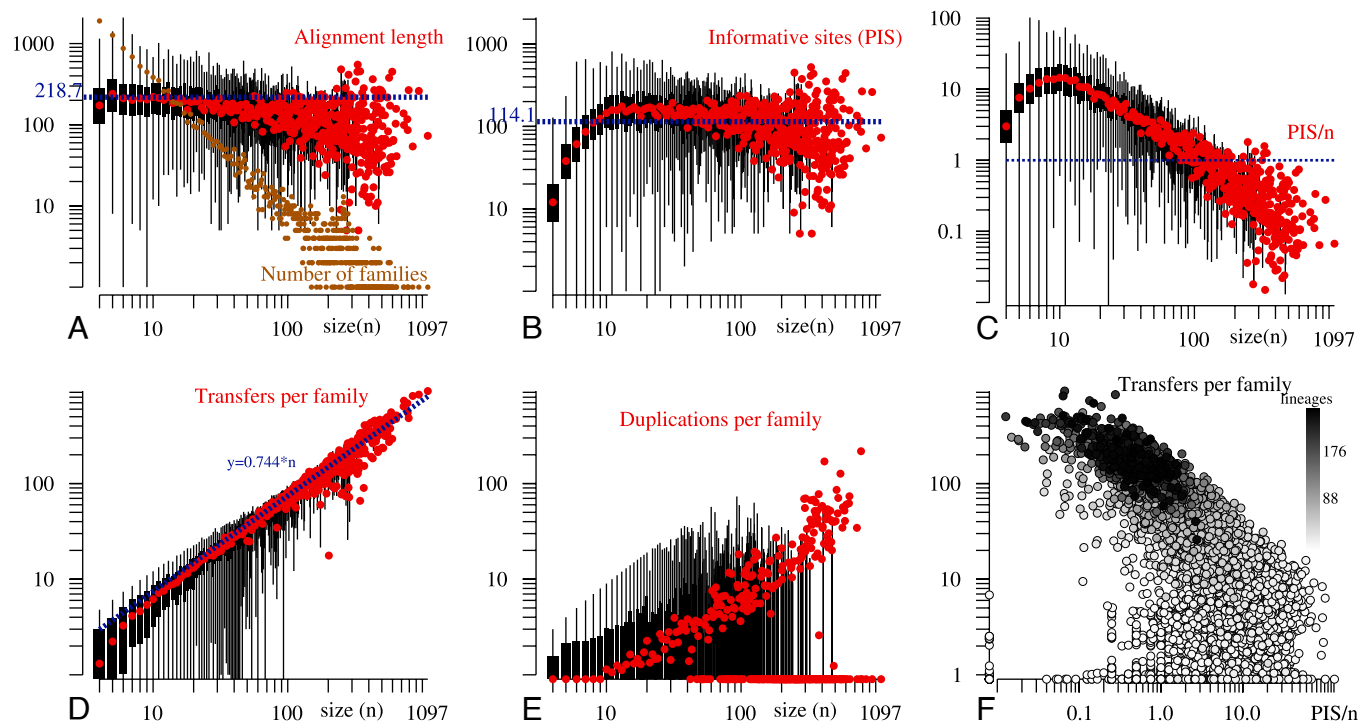


Fig. 5. Alignment and reconciliation statistics for a 265-genome reconstruction from ref. 19. On A–E, box-and-whisker plots show the distribution (interquartile range, minimum, and maximum; red circles denote medians) across families with same profile size on log-log scale. (A) Distribution of alignment length (number of sites after trimming and filtering) and number of families in function of alignment size n (copies in extant genomes). (B) Distribution of parsimony informative sites (PIS) by n . (C) Distribution of informative sites per sequence (PIS/ n). (D) Distribution of the number transfers per family; the dotted blue line denotes linear regression. (E) Distribution of the number of duplications per family. (F) Transfers and per-sequence informative sites for all families. Circles are shaded by family frequency across lineages (darker for shell and core with thresholds at 88 and 176 genomes as shown by the inset scale).

A similar balance is observed in P75 and E114 reconstructions (SI Appendix, section B.8).

These results collectively suggest that in archaeal evolution, a pervasive influx of rare genes is balanced by widespread loss, while a few lineages undergo massive expansions that define major lifestyle transitions.

Discussion

We have presented a numerical framework for the probabilistic inference of copy evolution in a phylogenetic GLD model. The model combines vertical (loss and duplication) and nonvertical (gain) processes of gene family evolution and thus can serve as a null model for hypothesizing about ancestral genomes, gene family histories, and evolutionary dynamics. Our experiments with a dataset of 269 Archaea show that the developed algorithms are suitable for inference over large and diverse sets of microbial genomes. The GLD inference may be swayed by the multitude of transient genes with just one or a few copies. As a remedy, we incorporated an adjustable observation bias into the inference, permitting the exclusion of halo families with too few members across extant genomes.

The main deviations in observed homolog families from the GLD model are caused by the assumption of (lineage- and family-wise) independent and (family-wise) identical processes. The model is thus equally useful for uncovering dependencies caused by natural selection. The family profiles show correlated lineage-specific losses across functionally linked families and across distant lineages. Many specific examples of functionally coupled genes exhibit lineage-specific patterns associated with lifestyle adaptations. Modular loss for metabolic pathways was equally observed in *phylogenetic profiling* of presence–absence

patterns (40), and, in turn, recurrent profiles reveal functional associations (41, 42).

Depending on existing genes from the same family, copies gained by horizontal gene transfer may be foundational for a gene family if there was no other copy before, or may constitute a fixed xenolog copy (a pseudoparalog) (11, 43, 44). The acquired xenolog may displace an existing copy. Xenologous gene displacements (XGD) are much less frequent than family gains among HGT events, which can be explained by the competitive advantage of the native gene (e.g., by codon bias), and applies in particular to the few core families that evolve primarily by inheritance (45). Probabilistic inference with the GLD model sums over all copy histories (and gene trees), considering family gains, XGD, and pseudoparalogs; even if the former are apparent and the latter two cannot be delineated from copy numbers alone.

The reconstructions from gene copies reinforce that Archaeal genome content is mainly shaped by gene loss, punctuated by large gains and duplications at some key ancestors (6, 46). For instance, massive increases of the gene repertoire are visible in the ancestors of the genus *Methanosarcina* (which possess three metabolic pathways for methanogenesis) and of the class *Halobacteria* (which are halophilic). Our simulations and inferences with the GLD model show a mathematical necessity of loss processes: that genes from the deepest ancestors often survive in extant genomes only infrequently, within shell and cloud families. Population-genetic models (47) and large studies (31) imply that genome streamlining is the prevalent mode of evolution among free-living prokaryotes because of large effective population sizes. Considering the halo families with one to three copies, however, we observe more balanced but widely dispersed pangenome dynamics with some large increases and an average equilibrium of genome size. The observation hints at the

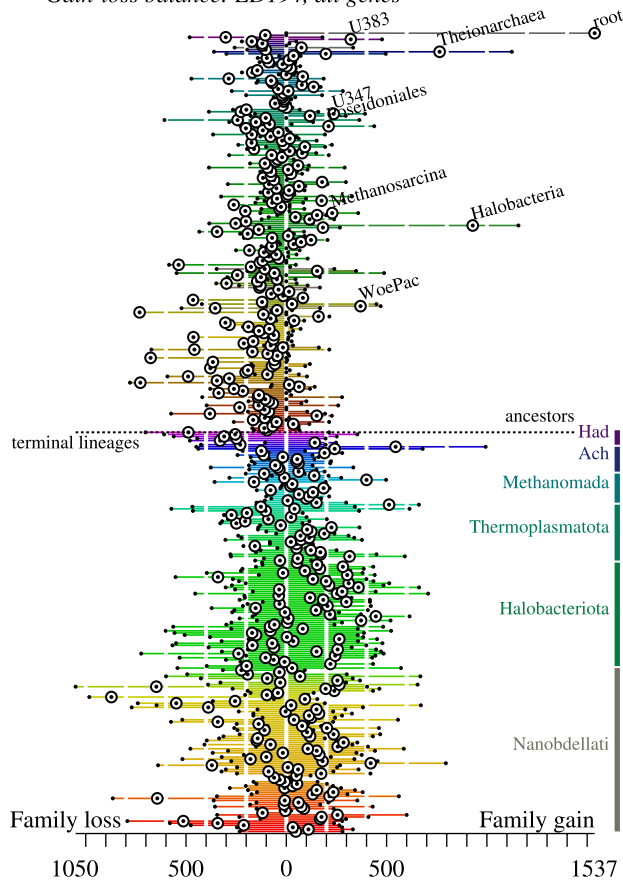


Fig. 6. Dynamics of family evolution in Methanobacteriati+Nanobdellati lineages (over all genes of ED194). Lineages are enumerated along the vertical axis, with terminal lineages in the lower half and ancestral lineages in the upper half. Horizontal bars to the left and right of 0 show posterior expectations for the number of families lost and gained on each branch. The node coloring and ordering follow Fig. 3. Dotted circles denote net change; largest increases are labeled: WoePac is the common ancestor of Woese and Pacearchaeales, U347 and U383 are low-height ancestors in MBG-D and Hadarchaeota.

pervasive arrival of diverse genes “in transit,” attaining fixation for a relatively short time (31) before submitting to elimination by selection. An implied immense pool of rare genes comprises unsampled prokaryotic lineages and the Archaeal virosphere (48); specific genes are available that facilitate even cross-domain transfers (39).

Our experiments on the reconstruction from simulated data show the tight accuracy of the maximum-likelihood inference, and the observed small distances from ground truth provide bounds on the maximum advantage gained from including gene sequences in the inference. Phylogenetic reconciliation methods consider the history of individual copies (gene trees) along the species tree by using the aligned gene sequences. ALE reconciliations (13) over very large phylogenies (19–21) infer rampant lateral gene transfer, especially in shell and core families, as well as implausibly few families inherited from deep ancestors.

Random heuristics, massive parallelization, and the lack of informative sites in the alignments highly influence the correctness and even the reproducibility of reconciliation studies (49). The close correlation of transfer events and family size in very large ALE reconstructions shows that gene trees are more incompatible with vertical inheritance as more sequences are aligned. Our analysis of alignment statistics suggests that inflated transfers are

statistical artifacts caused by inherently poor gene-tree sampling from too few informative sites.

Ancestral reconstructions from copy numbers avoid the computationally costly steps of multiple alignment and gene tree construction, which are increasingly error-prone and ambiguous with more sampled genomes. Our results show that maximum-likelihood inference with a GLD model can deliver statistically sound and biologically plausible hypotheses about the evolution of gene content covering the diversity of entire kingdoms.

Materials and Methods

Phylogenetic GLD Model. A phylogeny is a rooted binary tree with nodes numbered $u \in \{1, 2, \dots, R\}$: Every node either has two child nodes or is a terminal node (a leaf) with no children. The leaves are indexed as $\{1, 2, \dots, L\}$ and all indices respect the postfix order so that the last one is the root with index R . The tree is defined by its edges $T \subset [R] \times [R]$ directed from parent to child so that $u > v$ on every edge $uv \in T$.

Copy evolution for a homologous gene family is modeled by the joint distribution of the node copies $\{\xi_u\}_{u=1}^R$ that are nonnegative integer random variables associated with the nodes u in the phylogeny. The node copies are observable at the leaves (which correspond to extant species), and form the profile $\Xi = \{\xi_w\}_{w=1}^L$. Other nodes $L < v \leq R$ are (hypothetical) ancestors with unobserved copies ξ_v . On every edge uv , we insert the edge copies η_v between node copies ξ_u and ξ_v , which count the founding parental copies at the child. By definition $0 \leq \eta_v \leq \min\{\xi_u, \xi_v\}$. For uniformity, we also define $\eta_R = 0$ surely at the root, and a root prior that is the stationary distribution of a BDI process (arising thus technically from an infinitely long edge leading to the root). The phylogeny defines the conditional dependencies between the model variables giving the factorization for their joint distribution as

$$\begin{aligned} \mathbb{P}\{\xi_1 = n_1, \dots, \xi_R = n_R; \eta_1 = s_1, \dots, \eta_R = s_R = 0\} \\ = \mathbb{P}\{\xi_R = n_R \mid \eta_R = s_R\} \\ \times \prod_{uv \in T} \underbrace{\left(\mathbb{P}\{\eta_v = s_v \mid \xi_u = n_u\} \times \mathbb{P}\{\xi_v = n_v \mid \eta_v = s_v\}\right)}_{\text{deaths}} \underbrace{\left(\mathbb{P}\{\xi_v = n_v \mid \eta_v = s_v\}\right)}_{\text{births}}. \end{aligned} \quad [1]$$

The GLD model (7) posits that evolution along each edge $uv \in T$ follows a continuous-time BDI process parameterized by the edge length t_v , per-copy loss and duplication rates μ_v , λ_v , and the gain rate γ_v . For convenience, we assume a scaling with $\mu_v = 1$. Theorem 1 shows that the transient distributions between ξ , η in Eq. 1 amount to imposing that parent copies are lost independently by a loss parameter $0 \leq p_v \leq 1$ and additional copies for the child v are produced by a Pólya (negative binomial) or Poisson distribution with parameters (κ_v, q_v) or r_v where $0 \leq q_v \leq 1$ is the duplication parameter and $0 < r_v$ is the gain parameter.

Theorem 1. [Transient distributions] Let v be any node. Let $p_v = \frac{1 - e^{-(1-\lambda_v)t_v}}{1 - \lambda_v e^{-(1-\lambda_v)t_v}}$ if $\lambda_v \neq 1$, or $p_v = \frac{t_v}{1+t_v}$ if $\lambda_v = 1$. Define $q_v = \lambda_v p_v$, $r_v = \gamma_v p_v$, and if $\lambda_v \neq 0$, let $\kappa_v = r_v/q_v = \gamma_v/\lambda_v$. For all $0 \leq n$ and all $0 \leq s \leq n$, we have the probabilities

$$\begin{aligned} \mathbb{P}\{\eta_v = s \mid \xi_u = n\} &= \binom{n}{s} (1 - p_v)^s (p_v)^{n-s} & \text{if } uv \in T \\ \mathbb{P}\{\xi_v = n \mid \eta_v = s\} &= e^{-r_v} \frac{(r_v)^{n-s}}{(n-s)!} & \text{if } 0 = \lambda_v \\ \mathbb{P}\{\xi_v = n \mid \eta_v = s\} &= \binom{\kappa_v + n - 1}{n-s} (1 - q_v)^s q_v^{\kappa_v} (q_v)^{n-s} & \text{if } 0 \neq \lambda_v \end{aligned}$$

Proof: From Theorem 2 in ref. 7 and the ensuing discussion. $\square$

In a biologically meaningful model, it is reasonable to impose that the duplication rate does not exceed the loss rate, so $\lambda_u \leq 1$ or $q_v \leq p_v$ on every edge. Furthermore, we assume that the gain rate is also bounded as $\gamma_u \leq 1$ or

$r_v \leq p_v$ which is useful for regularizing the numerical computation of the gradient.

Since ξ_v and η_v are unbounded at ancestral nodes, the calculations are factorized by conditioning on the surviving node copies $\tilde{\xi}_v$ and edge copies $\tilde{\eta}_v$, which are progenitors of extant copies (SI Appendix, Fig. S1), thus necessarily bounded by the family size. The conditional distributions for surviving copies are expressed using the so-called survival parameters. Note that in contrast to Theorem 1, child edge copies jointly depend from the parent node copy with trinomial distribution for differential loss and shared inheritance.

Theorem 2. [Transition probabilities for surviving copies] Define the (per-copy) survival parameters $\tilde{p}_u, \tilde{q}_u$ and the (per-copy) extinction probability ϵ_u :

1. $\epsilon_u = 0$ for all leaves u , and for every nonleaf u with edges $uv, uw \in T$ leading to its children, $\epsilon_u = \tilde{p}_v \tilde{p}_w$.
2. At every node u , define $\tilde{p}_u$ and $\tilde{q}_u$ so that

$$\begin{aligned} \text{logit } \tilde{q}_u &= \ln \frac{\tilde{q}_u}{1 - \tilde{q}_u} = \text{logit } q_u + \ln(1 - \epsilon_u) \\ 1 - \tilde{p}_u &= (1 - p_u)(1 - \epsilon_u(1 - \tilde{q}_u)). \end{aligned} \quad [2]$$

3. At every node u with $\lambda_u = 0$, define $\tilde{r}_u = r_u(1 - \epsilon_u)$.

At every node v , and for all $0 \leq n$ and all $0 \leq s \leq n$,

$$\mathbb{P}\{\tilde{\xi}_v = n \mid \tilde{\eta}_v = s\} = e^{-\tilde{r}_v} \frac{(\tilde{r}_v)^{n-s}}{(n-s)!} \quad [3]$$

if $0 = \lambda_v$,
or

$$\mathbb{P}\{\tilde{\xi}_v = n \mid \tilde{\eta}_v = s\} = \binom{\kappa_v + n - 1}{n - s} (1 - \tilde{q}_v)^{s + \kappa_v} (\tilde{q}_v)^{n - s}$$

with $\kappa_v = \gamma_v / \lambda_v$ if $0 \neq \lambda_v$.

At every node u with child edges $uv, uw \in T$, for all $0 \leq n, 0 \leq s \leq n$ and $0 \leq t \leq n$ with $n \leq s + t$,

$$\begin{aligned} \mathbb{P}\{\tilde{\eta}_v = s, \tilde{\eta}_w = t \mid \tilde{\xi}_u = n\} &= \frac{n!}{(n-t)!(s+t-n)!(n-s)!} \\ &\times ((1 - p_v^-) \tilde{p}_w)^{n-t} \times ((1 - p_v^-)(1 - \tilde{p}_w))^s \times (p_v^-)^{n-s} \end{aligned}$$

with $\text{logit } p_v^- = \text{logit } \tilde{p}_v + \ln(1 - \tilde{p}_w)$.

Proof: Follows Theorem 5 in ref. 7, which defines $\tilde{q}_u = q_u \frac{1 - \epsilon_u}{1 - q_u \epsilon_u}$, $\tilde{p}_u = p_u + (1 - p_u) \epsilon_u \frac{1 - q_u}{1 - q_u \epsilon_u}$, and we define $p_v^- = \tilde{p}_v \frac{1 - \tilde{p}_w}{1 - \tilde{p}_v \tilde{p}_w}$. $\square$

Numerical Procedures. For the purposes of unconstrained optimization, the model is parameterized using logit and logarithmic scales. In particular, our parameters for numerical optimization are $\text{logit } p_u = \ln \frac{p_u}{1 - p_u}$, $\text{logit } \lambda_u$ and $\ln \gamma_u$ (or $\text{logit } \gamma_u$ with bounded gain rate). The calculations rely on logit- and log-domain arithmetics with our double-precision elementary operations such as ladd for addition on log-scale, limul for multiplication on logit-scale, and linv for conversion from logit- to log-scale: $\text{ladd}(\ln a, \ln b) = \ln(a + b)$, $\text{limul}(\text{logit } a, \text{logit } b) = \text{logit}(ab)$, and $\text{linv}(\text{logit } a) = (\ln a, \ln(1 - a))$. For instance, using Theorem 2, we have $\text{logit } \tilde{p}_v = -\text{limul}(-\text{logit } p_v, -\text{limul}(\text{logit } \epsilon_v, -\text{logit } \tilde{q}_v))$.

Likelihood Correction for Unobserved Profiles. Let $\Xi = \{\xi_v = n_v \mid 1 \leq v \leq L\}$ be an observed profile at the leaves. We are interested in computing the likelihood of Ξ which is the infinite summation of the terms from Eq. 1 across the possible values of the unobserved random variables

$$\begin{aligned} L(\Xi) &= \mathbb{P} \Xi \\ &= \sum_{n_R} \sum_{s_{R-1}} \sum_{n_{R-1}} \cdots \sum_{s_{L+1}} \sum_{n_{L+1}} \mathbb{P}\{\xi_v = n_v\}_{v=1}^R, \{\eta_v = s_v\}_{v=1}^R \}, \end{aligned}$$

where node copies take integer values $0 \leq n_u$ at the ancestral nodes $L + 1 \leq u \leq R$, edge copies take integer values $0 \leq s_v \leq \min\{n_u, n_v\}$ on every edge $uv \in T$, and $s_R = 0$ at the root.

The distribution parameters are fitted to a sample of random profiles $\mathcal{S} = \{\Xi^1, \dots, \Xi^F\}$ observed for F families with independent and identically distributed (iid) joint copy distributions $\{(\eta_{f,u}, \xi_{f,u})\}_{u=1}^R$ for $f = 1, \dots, F$. By independence, the (uncorrected) sample log-likelihood is $\ln L = \sum_{f=1}^F \ln L(\Xi^f)$.

The profile size is the sum of observed copies: $\Omega(\Xi) = \sum_{w=1}^L \xi_w$. In practice, the families are observed only if they have some members detected at the leaves: at least $\Omega_{\min} = 1$, but, if gene families are inferred from alignments, then at least $\Omega_{\min} = 2$ members are required. Datasets for phylogenetic reconciliation have $\Omega_{\min} = 4$. The sampling bias must be corrected in the log-likelihood (50): set

$$L^*(\Xi) = \mathbb{P}\{\Xi \mid \Omega_{\min} \leq \Omega(\Xi)\} = \frac{L(\Xi)}{\mathbb{P}\{\Omega_{\min} \leq \Omega(\Xi)\}}$$

for an observed profile Ξ large enough to pass the observation threshold $\Omega_{\min}$. The corrected log-likelihood for the sample is thus

$$\ln L^* = \sum_{f=1}^F \ln L(\Xi^f) - F \ln(1 - L(0)), \quad [4]$$

where Ξ^f are the observed family profiles and $L(0) = \mathbb{P}\{\Omega(\Xi) < \Omega_{\min}\}$ is our short-hand notation for the likelihood of an unobserved profile.

For any model parameter θ , the partial derivatives of the corrected log-likelihood are calculated by Eq. 4 as

$$\frac{\partial}{\partial \theta} \ln L^* = \sum_{f=1}^F \frac{\partial}{\partial \theta} \ln L(\Xi^f) + F \frac{L(0)}{1 - L(0)} \frac{\partial}{\partial \theta} \ln L(0). \quad [5]$$

Likelihood and Gradient for Observed Profiles. The computations over a profile can be factorized along the dependencies between surviving edge and node copies from Theorem 2. We used the techniques of Csürös (7) to develop the computations with only summation and multiplication (in log-scaled representation) for the profile log-likelihoods $\ln L(\Xi^f)$ in Eq. 4, its partial derivatives in Eq. 5, and the posterior expectations. In order to avoid loss of precision, the signed gradient components are represented and manipulated as log-scaled differences of the form $(e^a - e^b)$, so that only a single subtraction is needed at the end for evaluation. See SI Appendix, section A for the calculations. The main mathematical ideas are analogous to the computations for unobserved profiles (Theorems 3–5).

Likelihood and Posteriors for Unobserved Profiles. Define the random sums of leaf copies at all nodes u by $\Omega_u = \sum_{v \in \mathcal{L}_u} \xi_v$ where $\mathcal{L}_u$ denotes the set of leaves in the subtree rooted at u . Define the complementary copy sums $\Omega_u^* = \Omega_R - \Omega_u = \sum_{v \in \mathcal{L} - \mathcal{L}_u} \xi_v$. Let $M = \Omega_{\min} - 1$ denote the maximum size of an unobserved family. We want to compute the likelihood of an unobserved profile $L(0) = \mathbb{P}\{\Omega_R \leq M\}$ where Ω_R is the profile size. Define the inner conditional log-likelihoods

$$\begin{aligned} \tilde{K}_{u,m}(s) &= \ln \mathbb{P}\{\Omega_u = m \mid \tilde{\eta}_u = s\} \\ \tilde{C}_{u,m}(n) &= \ln \mathbb{P}\{\Omega_u = m \mid \tilde{\xi}_u = n\} \end{aligned}$$

at all nodes $1 \leq u \leq R$, and $0 \leq s \leq m$, $0 \leq n \leq m$ and $0 \leq m \leq M$. Define the complementary outer log-priors

$$\tilde{B}_{u,k}(n) = \ln \mathbb{P}\{\Omega_u^* = k; \tilde{\xi}_u = n\} \quad \tilde{J}_{u,k}(s) = \ln \mathbb{P}\{\Omega_u^* = k; \tilde{\eta}_u = s\}.$$

Theorem 3. [Unobserved survival likelihood recurrences] At any node u ,

$$\ln L(0) = \ln \sum_{m=0}^M \sum_{k=0}^{M-m} \sum_{n=0}^m \exp(\tilde{C}_{u,m}(n) + \tilde{B}_{u,k}(n)).$$

We have the following recurrences

- At every node u : for all $m < s$, $\tilde{\mathbf{K}}_{u,m}(s) = -\infty$ and for all $m < n$, $\tilde{\mathbf{C}}_{u,m}(n) = -\infty$.
- At every node u , for all $0 \leq s \leq m \leq M$,

$$\tilde{\mathbf{K}}_{u,m}(s) = \ln \sum_{n=s}^m \exp(\tilde{\mathbf{K}}'_{u,m}(n, s)) \quad \text{with}$$

$$\tilde{\mathbf{K}}'_{u,m}(n, s) = \tilde{\mathbf{C}}_{u,m}(n) + \ln \mathbb{P}\{\tilde{\xi}_u = n \mid \tilde{\eta}_u = s\}.$$

- If u is a leaf, then $\tilde{\mathbf{C}}_{u,m}(n) = -\infty$ when $n \neq m$ and $\tilde{\mathbf{C}}_{u,m}(m) = 0$. If u is an inner node with child edges $uv, uw \in T$, then for all $0 \leq n \leq m \leq M$,

$$\begin{aligned} \tilde{\mathbf{C}}_{u,m}(n) = \ln \sum_{s=0}^n \left(\sum_{k=n-s}^{m-s} \exp(\tilde{\mathbf{K}}_{v,m-k}(s) + \tilde{\mathbf{L}}_{w,k}(n, n-s)) \right) \\ \times \binom{n}{s} (1 - p_v^-)^s (p_v^-)^{n-s} \end{aligned}$$

with $\tilde{\mathbf{L}}_{w,k}(n, n) = \tilde{\mathbf{K}}_{w,k}(n)$, and, for all $0 \leq t < n$,

$$\begin{aligned} \tilde{\mathbf{L}}_{w,k}(n, t) = \ln \left((1 - \tilde{p}_w) \times \exp(\tilde{\mathbf{L}}_{w,k}(n, t+1)) \right. \\ \left. + \tilde{p}_w \times \exp(\tilde{\mathbf{L}}_{w,k}(n-1, t)) \right). \end{aligned}$$

- At the root, $\tilde{\mathbf{J}}_{R,k}(0) = 0$ and $\tilde{\mathbf{J}}_{R,k}(s) = -\infty$ for all $0 < s \leq M$ and for all $0 \leq k \leq M$.
- At every nonroot node v with parent u and sibling w (i.e., $uv, uw \in T$), for all $0 \leq s \leq M$ and $0 \leq k \leq M$,

$$\begin{aligned} \tilde{\mathbf{J}}_{v,k}(s) = \ln \sum_{n=s}^M \exp(\tilde{\mathbf{J}}'_{v,k}(n, s)) \quad \text{with} \\ \tilde{\mathbf{J}}'_{v,k}(n, s) = \ln \sum_{j=n-s}^k \exp(\tilde{\mathbf{B}}_{u,k-j}(n) + \tilde{\mathbf{L}}_{w,j}(n, n-s)) \\ \times \binom{n}{s} (1 - p_v^-)^s + (p_v^-)^{n-s} \end{aligned}$$

- At every node u , for all $0 \leq n \leq M$ and $0 \leq k \leq M$

$$\begin{aligned} \tilde{\mathbf{B}}_{u,k}(n) = \ln \sum_{s=0}^n \exp(\tilde{\mathbf{B}}'_{u,k}(n, s)) \quad \text{with} \\ \tilde{\mathbf{B}}'_{u,k}(n, s) = \tilde{\mathbf{J}}_{u,k}(s) + \ln \mathbb{P}\{\tilde{\xi}_u = n \mid \tilde{\eta}_u = s\} \end{aligned}$$

Proof: See SI Appendix, section A. $\square$

We have the log-likelihood and the posterior probabilities from the conditional likelihoods of Theorem 3.

Theorem 4. [Posteriors for unobserved profiles] We have the posteriors at nodes u and edges uv

$$\begin{aligned} \mathbb{P}\{\tilde{\eta}_u = s, \tilde{\xi}_u = n \mid \Omega_R \leq M\} \\ = \sum_{m=n}^M \sum_{k=0}^{M-m} \exp(\tilde{\mathbf{B}}'_{u,k}(n, s) + \tilde{\mathbf{C}}_{u,m}(n) - \ln L(0)), \end{aligned}$$

$$\begin{aligned} \mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_v = s \mid \Omega_R \leq M\} \\ = \sum_{m=s}^M \sum_{k=0}^{M-m} \exp(\tilde{\mathbf{J}}'_{v,k}(n, s) + \tilde{\mathbf{K}}_{v,m}(s) - \ln L(0)), \end{aligned}$$

$$\begin{aligned} \mathbb{P}\{\tilde{\eta}_v = s \mid \Omega_R \leq M\} \\ = \sum_{m=s}^M \sum_{k=0}^{M-m} \exp(\tilde{\mathbf{J}}_{v,k}(s) + \tilde{\mathbf{K}}_{v,m}(s) - \ln L(0)). \end{aligned}$$

Proof: See SI Appendix, section A. $\square$

Gradient Calculations. We arrive at the gradient of the corrected log-likelihood in three steps. First, we calculate the gradient with respect to the survival parameters $\text{logit } \tilde{p}_u$, $\text{logit } \tilde{q}_u$ and $\ln \kappa_u$ or $\ln \tilde{r}_u$. Second, the survival gradient is transformed into partial derivatives by the GLD distribution parameters $\text{logit } p_u$, $\text{logit } q_u$ and $\ln \kappa_u$ or $\ln r_u$, using a preorder traversal of the phylogeny that "unfolds" the extinctions toward the leaves (Theorem 6). Third, we compute the gradient with respect to the duplication and gain rates $\text{logit } \lambda$ and $\ln \gamma$ using the chain rule (Theorem 7). For a function Φ (in particular, for log-likelihoods), we denote the partial derivatives by $\Phi^{(\theta)} = \frac{\partial \Phi}{\partial \theta}$, and the column vector of partial derivatives by $\nabla \Phi$.

It follows from Theorem 3 that the survival-parameter gradient for unobserved profiles can be calculated from posterior probabilities of Theorem 4 as shown in Theorem 5.

Theorem 5. [Gradient for unobserved profiles] Let $\Phi = \ln L(0) = \ln \mathbb{P}\{\Omega_R \leq M\}$. Define the tail probabilities for births at every node u

$$\delta_{u,i}^+ = \ln \left(\mathbb{P}\{i < \tilde{\xi}_u \mid \Omega_R \leq M\} - \mathbb{P}\{i < \tilde{\eta}_u \mid \Omega_R \leq M\} \right)$$

and the tail probabilities for deaths on every edge

$$uv\delta_{v,i}^- = \ln \left(\mathbb{P}\{i < \tilde{\xi}_u \mid \Omega_R \leq M\} - \mathbb{P}\{i < \tilde{\eta}_v \mid \Omega_R \leq M\} \right)$$

Define the expectations

$$\sigma_u = \ln \mathbb{E}[\tilde{\eta}_u \mid \Omega_R \leq M] = \ln \sum_{i=0}^M \ln \mathbb{P}\{i < \tilde{\eta}_u \mid \Omega_R \leq M\}$$

$$\delta_u^+ = \ln \mathbb{E}[\tilde{\xi}_u - \tilde{\eta}_u \mid \Omega_R \leq M] = \ln \sum_{i=0}^M \exp(\delta_{u,i}^+)$$

$$\delta_v^- = \ln \mathbb{E}[\tilde{\xi}_u - \tilde{\eta}_v \mid \Omega_R \leq M] = \ln \sum_{i=0}^M \exp(\delta_{v,i}^-).$$

- At every node $1 \leq u \leq R$,

$$\delta_{u,i}^+ = \ln \sum_{n=i+1}^M \sum_{s=0}^i \mathbb{P}\{\tilde{\eta}_u = s, \tilde{\xi}_u = n \mid \Omega_R \leq M\}$$

If $0 < q_u$, then

$$\Phi^{(\text{logit } \tilde{q}_u)} = (1 - \tilde{q}_u) \exp(\delta_u^+) - \tilde{q}_u \exp(\text{ladd}(\sigma_u, \ln(\kappa_u)))$$

$$\Phi^{(\ln \kappa_u)} = \left(\sum_{0 \leq i} \frac{\kappa_u}{\kappa_u + i} \exp(\delta_{u,i}^+) \right) - \kappa_u (-\ln(1 - \tilde{q}_u)),$$

and, if $q_u = 0$, then

$$\Phi^{(\ln \tilde{r}_u)} = \exp(\delta_u^+) - \tilde{r}_u.$$

- On every edge $uv \in T$,

$$\delta_{v,i}^- = \ln \sum_{n=i+1}^M \sum_{s=0}^i \mathbb{P}\{\tilde{\xi}_u = n, \tilde{\eta}_v = s \mid \Omega_R \leq M\}$$

$$\Phi^{(\text{logit } \tilde{p}_v)} = (1 - p_v^-) \exp(\delta_v^-) - p_v^- \exp(\sigma_v).$$

Proof: See SI Appendix, section A. $\square$

The gradient by the survival parameters is transformed into the gradient by original model parameters as shown by the following *Theorem*, applicable to any function Φ (we apply it to $\Phi = \ln L^*$).

Theorem 6. [Exact gradient]

1. At the root R , if $q_R \neq 0$, then

$$\nabla \Phi(\text{logit } q_R, \text{logit } \epsilon_R) = \begin{pmatrix} \Phi(\text{logit } \tilde{q}_R) \\ -\epsilon_R \Phi(\text{logit } \tilde{q}_R) \end{pmatrix};$$

and if

$$q_R = 0,$$

then

$$\nabla \Phi(\ln r_R, \text{logit } \epsilon_R) = \begin{pmatrix} \Phi(\ln \tilde{r}_R) \\ -\epsilon_R \Phi(\ln \tilde{r}_R) \end{pmatrix}.$$

2. Let $v < R$ be any nonroot node with parent u . If $q_v \neq 0$, then

$$\begin{aligned} \nabla^T \Phi(\text{logit } p_v, \text{logit } q_v, \text{logit } \epsilon_v) \\ = \nabla^T \Phi(\text{logit } \tilde{p}_v, \text{logit } \tilde{q}_v, \text{logit } \epsilon_u) \\ \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 - p_v^- & 0 \end{pmatrix} \begin{pmatrix} \frac{p_v}{\tilde{p}_v} & -q_v \frac{\epsilon_v(1 - \tilde{q}_v)}{\tilde{p}_v} & \frac{\epsilon_v(1 - \tilde{q}_v)}{\tilde{p}_v} \\ 0 & 1 & -\epsilon_v \end{pmatrix} \end{aligned}$$

If $q_v = 0$, then

$$\begin{aligned} \nabla^T \Phi(\text{logit } p_v, \text{logit } \epsilon_v, \ln r_v) = \nabla^T \Phi(\text{logit } \tilde{p}_v, \text{logit } \epsilon_u, \ln \tilde{r}_v) \\ \times \begin{pmatrix} 1 & 0 \\ 1 - p_v^- & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{p_v}{\tilde{p}_v} & \frac{\epsilon_v}{\tilde{p}_v} & 0 \\ 0 & -\epsilon_v & 1 \end{pmatrix} \end{aligned}$$

Proof: See *SI Appendix, section A*. $\square$

The last step is to first convert into gradients by the log-gain rate $\ln \gamma$ and then by $\text{logit } \lambda$ (*Theorem 7*).

Theorem 7. [Gradient by gain and duplication rates] For every node $1 \leq v \leq R$, if $q_v \neq 0$, then

$$\begin{aligned} \nabla^T \Phi(\text{logit } p_v, \text{logit } q_v, \ln \gamma_v) = \nabla^T \Phi(\text{logit } p_v, \text{logit } q_v, \ln \kappa_v) \\ \times \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 - p_v & -(1 - q_v) & 1 \end{pmatrix}; \end{aligned}$$

and if

$$q_v = 0,$$

then

$$\nabla^T \Phi(\text{logit } p_v, \ln \gamma_v) = \nabla^T \Phi(\text{logit } p_v, \ln r_v) \times \begin{pmatrix} 1 & 0 \\ 1 - p_v & 1 \end{pmatrix}.$$

For a duplication rate $\lambda_v = q_v/p_v \leq 1$, we have

$$\begin{aligned} \nabla^T \Phi(\text{logit } p_v, \text{logit } \lambda_v) = \nabla^T \Phi(\text{logit } p_v, \text{logit } q_v) \\ \times \begin{pmatrix} 1 & 0 \\ \frac{1 - p_v}{1 - q_v} & \frac{1 - \lambda_v}{1 - q_v} \end{pmatrix} \end{aligned}$$

Proof: See *SI Appendix, section A*. $\square$

Algorithmic Complexity and Likelihood Maximization. The likelihood correction takes a quartic time.

Theorem 8. [Running time and memory for unobserved profiles] The likelihood and the gradient for unobserved profiles of size at most M can be computed in $\Theta(LM^4)$ time using $\Theta(LM^3)$ memory, where L is the number of terminal taxa.

Proof: See *SI Appendix, section A*. $\square$

Computing the posteriors and the gradient for one profile takes, at each node u , a quadratic time in the profile size $\Omega_u = \Omega(\Xi_u)$.

Theorem 9. The exact computation of the sample likelihood and its gradient with bias correction $\Omega_{\min} = o((FL)^{1/4})$ takes $\Theta(FhL^2\bar{\omega}^2)$ time where

$$\bar{\omega} = \sqrt{\frac{1}{F} \sum_{f=1}^F \left(\frac{\Omega(\Xi^f)}{L} \right)^2}$$

is the quadratic average of the mean copies across the sample, h is the height of the tree, and L is the number of leaves.

Proof: See *SI Appendix, section A*. $\square$

The exact gradient computation can be used with a general-purpose numerical optimization algorithm. In particular, we implemented the quasi-Newton method of Broyden-Fletcher-Goldfarb-Shanno (Algorithms 3.5, 3.6, and 6.1 in ref. 22) with the line-search method of Moré and Thuente (51).

Datasets and analysis methods. We identified all genome assemblies from ref. 24 that had unique NCBI Taxonomy identifiers and complete annotations in NCBI GenBank. We downloaded the GFF annotations and protein Fasta files, either from RefSeq (52) or from GenBank assemblies, preferring the former if it was available. We ended up with 269 genomes and their gene repertoires. We compiled a BLAST (53) database over domain occurrences from the arCOG (25) and asCOG (26) collections. We excluded sequences with less than 60 amino acids and assigned the remaining proteins in our 269 genomes to arCOG families using psiblast (54) with E-value cutoff of 10^{-9} . The sequences that did not match any arCOG occurrences were assigned to asCOG families using psiblast in the same way. We grouped the remaining sequences that did not match any arCOG or asCOG families using a standard Markov-clustering pipeline (55) with inflation parameter 1.2 over all-against-all BLAST search results. In a final pass, we added all genes with identical protein-coding sequences by parsing the GFF annotation files with custom Perl scripts.

ML optimizations, ancestral reconstructions, simulations, and various filtering steps were done using the Count (23) software package implementing the described algorithms.

We modified the phylogeny from ref. 24 subsampled over our 269 genomes to reflect the branching order of phyla in ref. 27, using Count for making the necessary subtree prune-and-regraft operations by hand. We established the naming of major clades following the Genome Taxonomy Database (56) and (57).

Datasets for the ALE reconciliations were extracted from the supporting material provided by the authors as indicated for the references (17–21). For the benchmarking of phylogenetic incongruence with short alignments, we used IQ-TREE3 (58) to simulate alignments with the model from ref. 19 and to sample gene trees following the protocol of the same study.

Data visualization was coded in PostScript.

Data, Materials, and Software Availability. The described algorithms are implemented in the Count software package, which can be downloaded from <https://github.com/csurosm/count/> in Java source code or as executable jar file (23). Data (269-genome archaeal dataset and Count analysis sessions) have been deposited in Zenodo at <https://doi.org/10.5281/zenodo.18071341> (59).

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